

Customer Shopping Behavior Analysis

3,900 transactions · 18 features · Python · SQL · Power BI





Dataset at a Glance

3,900

Transactions

18

Features

37

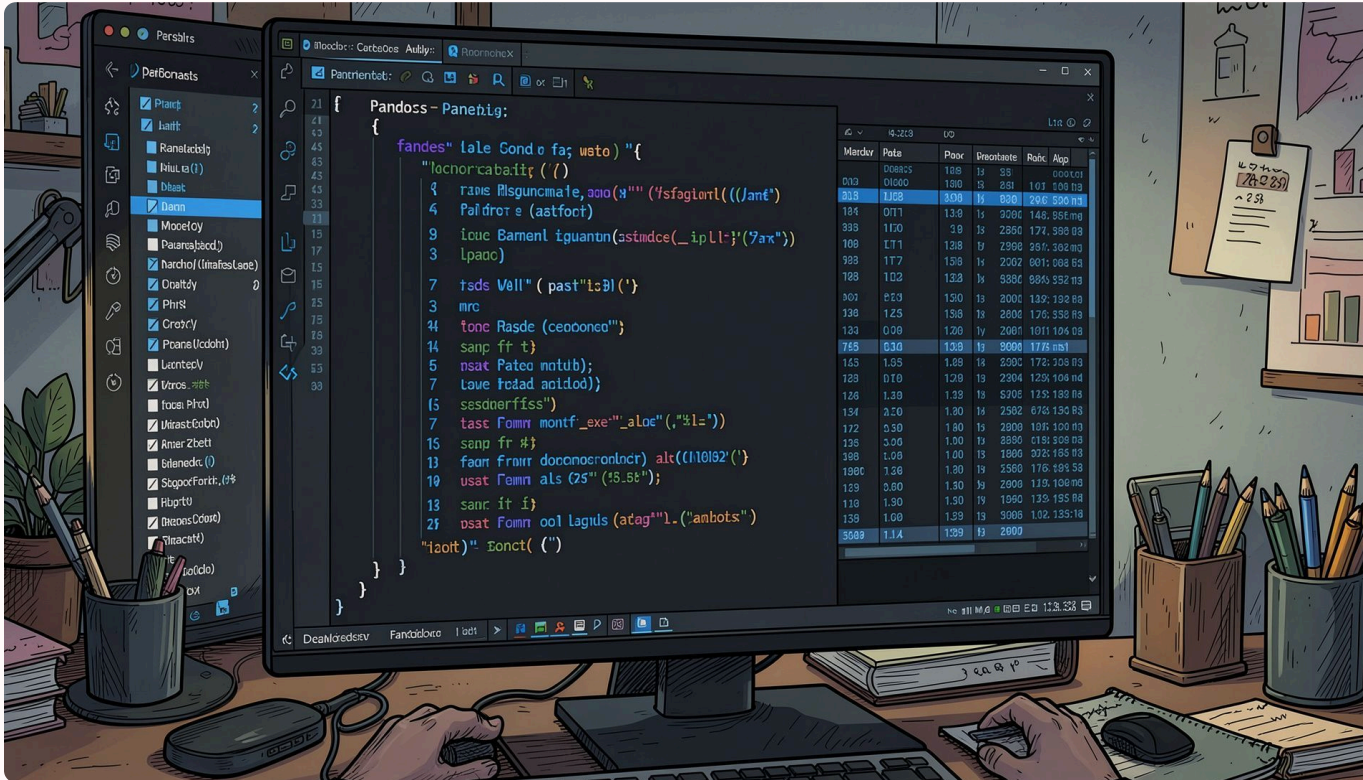
Missing Values

Review Rating only

\$59.76

Avg Purchase

Data Preparation in Python



→ Clean & Standardize

Snake_case columns, median imputation for Review Rating

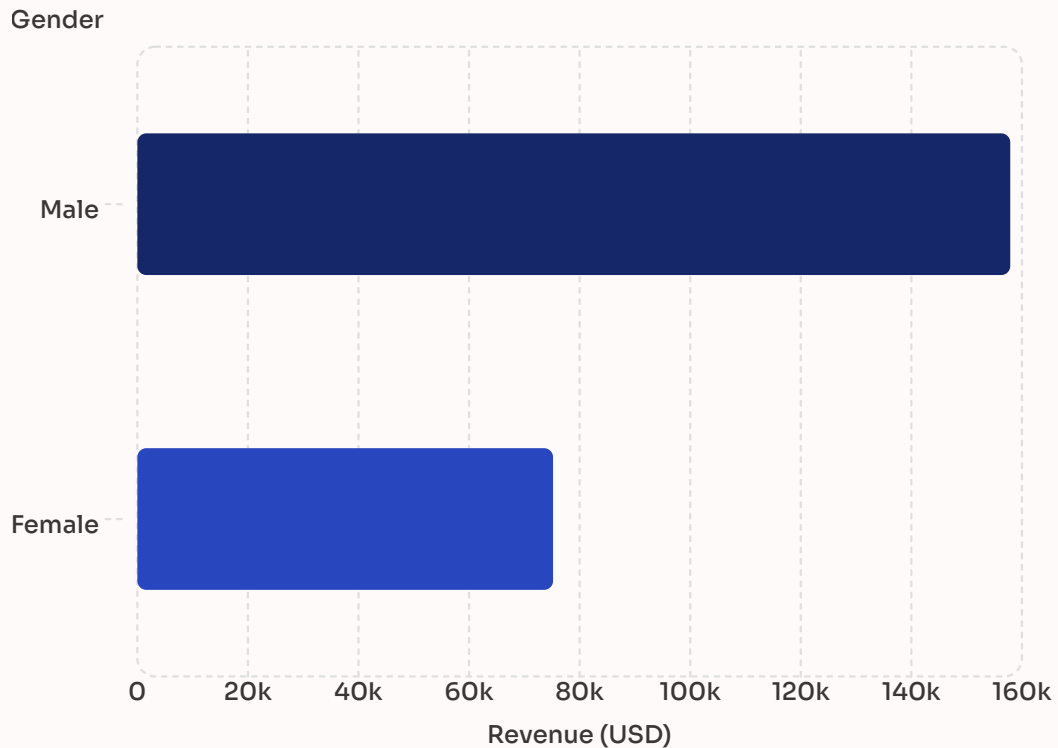
→ Feature Engineering

Created `age_group` and `purchase_frequency_days`

→ Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis

Revenue & Spending Insights



Key Findings

- **Male customers** generated 2× more revenue than females
- Discount users still spent **above average** (\$62–\$97)
- Express shipping avg: **\$60.48** vs Standard **\$58.46**

Top Products by Rating

 **Gloves**

3.86 avg rating

 **Sandals**

3.84 avg rating

 **Boots**

3.82 avg rating

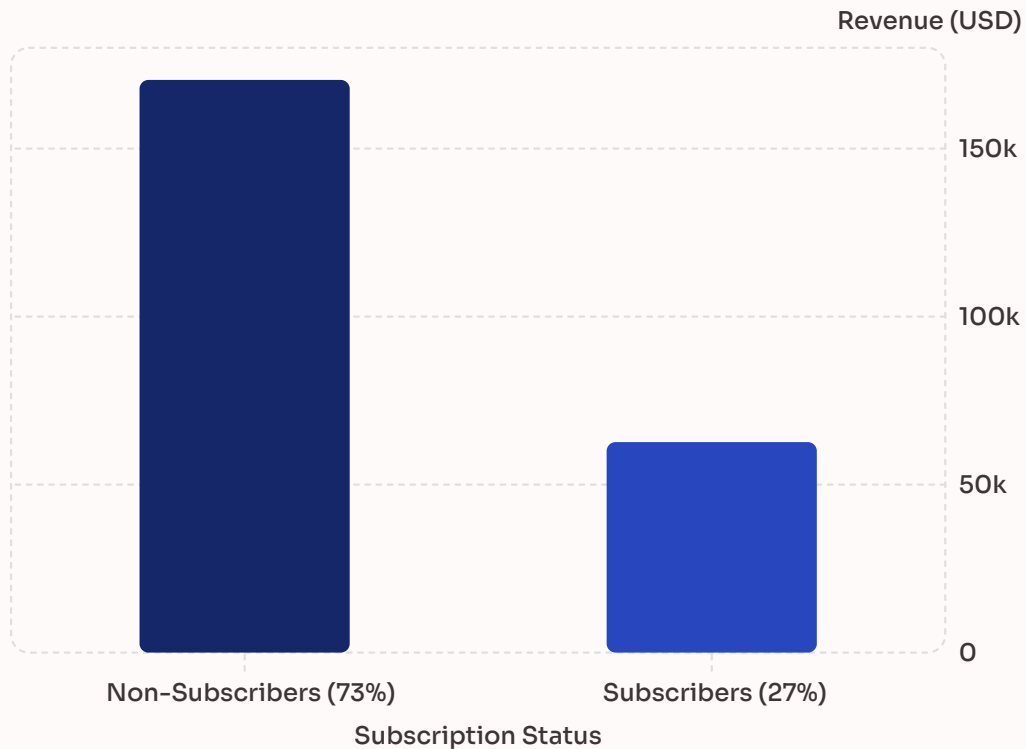
Hat

3.80 avg rating

Skirt

3.79 avg rating

Subscribers vs Non-Subscribers

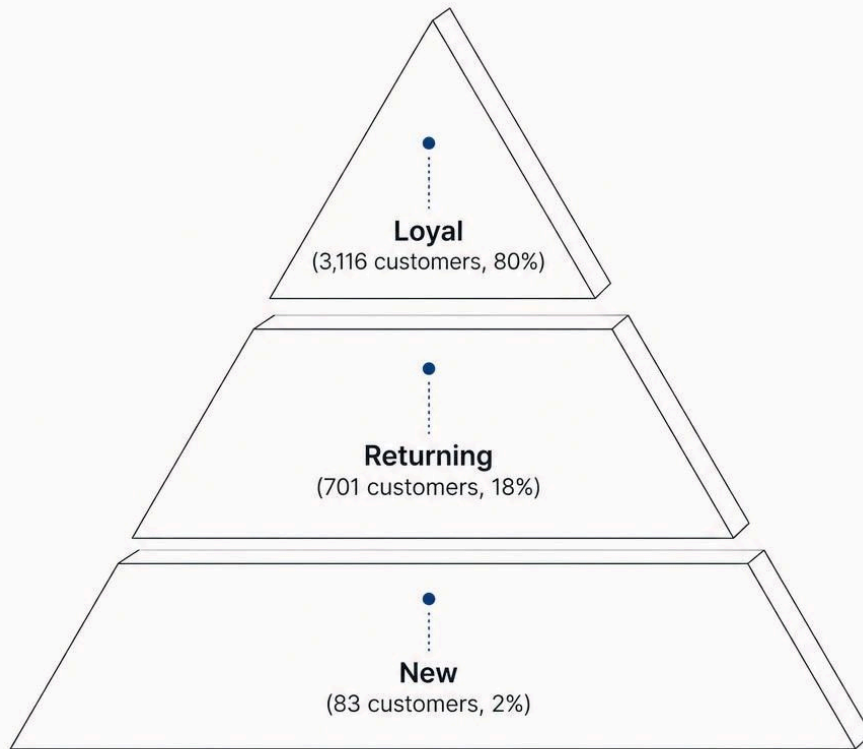


The Gap

73% of customers are **non-subscribers**, yet average spend is nearly identical (~\$59.50).

📌 **Opportunity:** 958 loyal repeat buyers already subscribe — convert the remaining 2,518.

Customer Segmentation

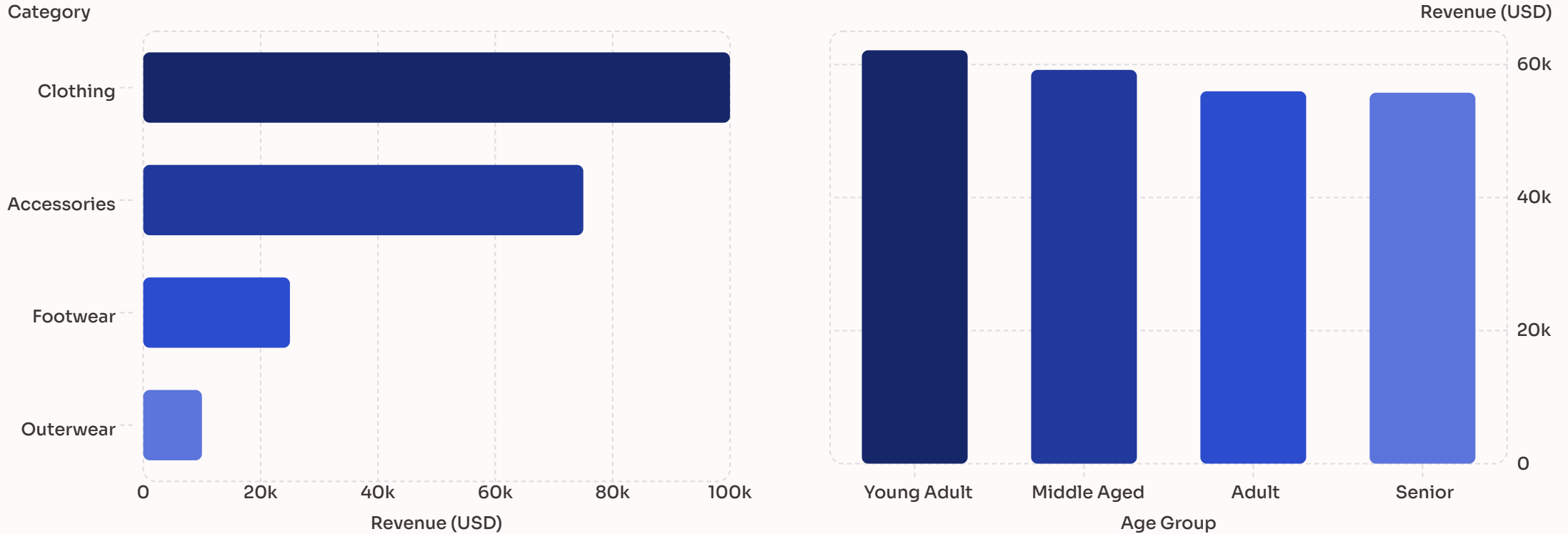


Segment Breakdown

- **Loyal:** 3,116 customers — 80% of base
- **Returning:** 701 customers — 18%
- **New:** 83 customers — only 2%

Strong loyalty base, but new customer acquisition needs attention.

Revenue by Category & Age Group



Clothing dominates revenue. Young Adults lead all age groups.



Power BI Dashboard

3.9K Customers

\$59.76 Avg Purchase

3.75 Avg Rating

Interactive filters: Subscription Status, Gender, Category, Shipping Type



Business Recommendations



Boost Subscriptions

Incentivize 73% non-subscribers



Grow Clothing & Accessories

Top revenue drivers — expand here



Target Young Adults

Highest revenue segment



Raise AOV

Cross-sell; free-shipping threshold above \$59.76